TableTalk Technical Test Report

Narrative Voice Processing and Classification

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1 Overview

This report summarizes the full pipeline run on a RAVDESS subset (200 speech recordings): (1) audio preprocessing and feature extraction, (2) narrative tone classification, (3) AI-based transcription, and (4) narrative audio retrieval simulation.

2 Dataset and Split

Data source: RAVDESS Emotional Speech Dataset (speech subset). A balanced subset of 200 clips was used (8 emotion classes, 25 clips per class), consistent with the assignment recommendation of using a subset (50–200 recordings).

Table 1: Dataset split summary

Split	Train	Validation	Test
Clip count	140	20	40

Table 2: Label distribution (balanced subset)

Label	Count
angry	25
calm	25
disgust	25
fearful	25
happy	25
neutral	25
sad	25
surprised	25

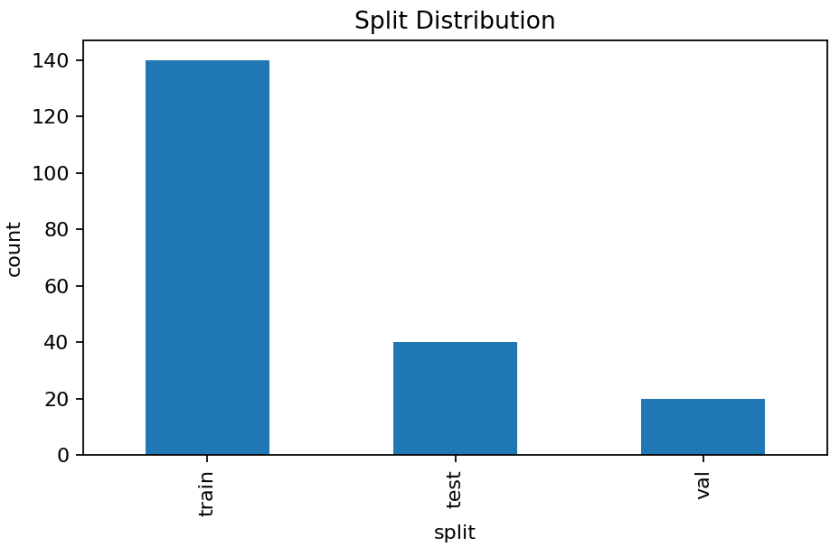

Figure 1: Split distribution

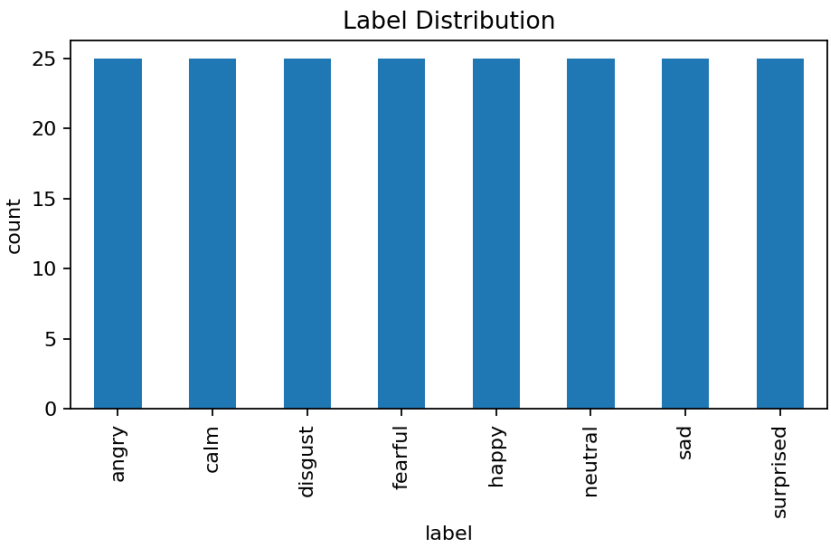

Figure 2: Label distribution

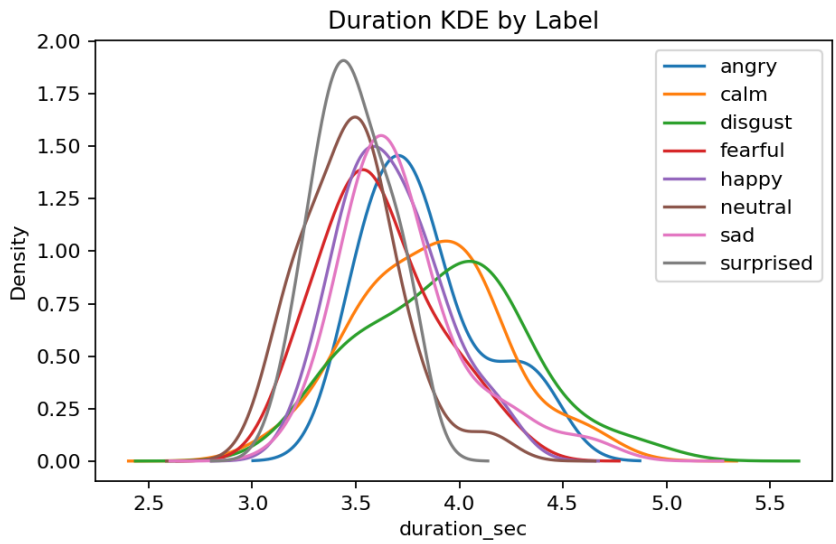

Figure 3: Duration KDE by label

3 Task 1: Audio Processing and Feature Engineering

Pipeline summary:

- Resampling to 16 kHz, mono conversion.
- Peak normalization.
- Feature extraction: duration, pitch statistics, spectral centroid statistics, energy statistics, and MFCC statistics.
- Structured output for modeling: one row per clip in `data/features/features.csv`.

4 Task 2: Narrative Tone Classification

Candidate models were Logistic Regression and Random Forest. Model selection on validation split:

Table 3: Validation model comparison (macro-F1)

Model	Validation macro-F1
Logistic Regression	0.3589
Random Forest	0.2524

Best model: **Logistic Regression**.

Table 4: Test metrics (best model)

Metric	Value
Accuracy	0.3500
Macro-F1	0.3427
Weighted-F1	0.3427

Table 5: Per-class test results

Class	Precision	Recall	F1
angry	0.5000	0.2000	0.2857
calm	0.5000	0.4000	0.4444
disgust	0.4000	0.4000	0.4000
fearful	0.4286	0.6000	0.5000
happy	0.0000	0.0000	0.0000
neutral	0.5000	0.4000	0.4444
sad	0.1429	0.2000	0.1667
surprised	0.4286	0.6000	0.5000

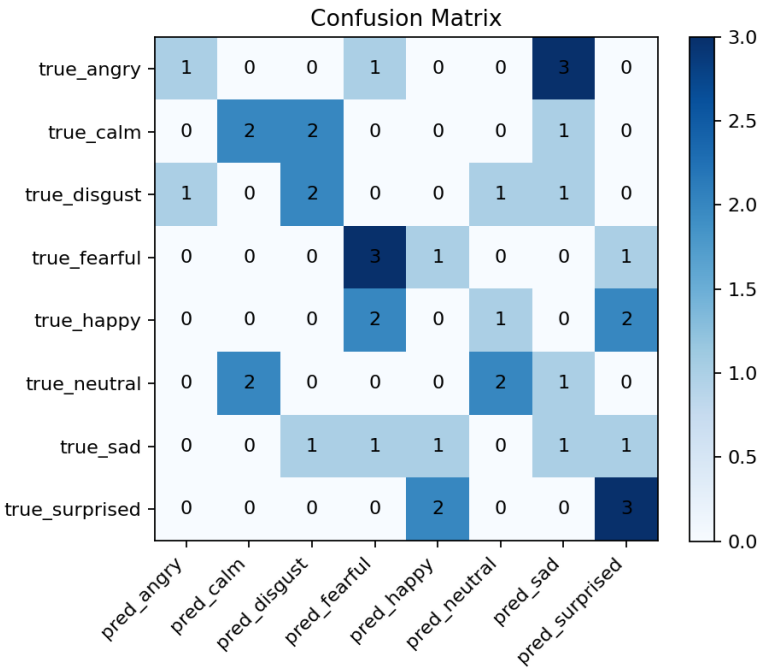

Figure 4: Confusion matrix on test split

Short discussion. The 8-class setup is non-trivial with handcrafted features and a small subset size. Current results are above random baseline ($1/8 = 12.5\%$) but still modest, indicating room for stronger representations (e.g., pretrained speech embeddings) and further tuning.

5 Task 3: AI-Based Transcription

ASR model: Whisper (`tiny`). A total of 200 clips were transcribed and evaluated against RAVDESS canonical statements.

Table 6: ASR evaluation

Metric	Value
Pairs evaluated	200
WER	0.0658
CER	0.0359

Short discussion. The transcription pipeline is stable on this subset (200/200 processed). The observed WER/CER indicates generally reliable recognition for the limited phrase set in RAVDESS.

6 Task 4: Narrative Audio Retrieval (TableTalk Simulation)

A rule-based retrieval prototype was implemented using label, duration, acoustic features, and transcript text.

Table 7: Example retrieval queries and returned results

Query	Hits
`"calm narration >4 seconds"`	9
`"calm narration >4 seconds"` (saved demo file)	10
`"high-energy speech"`	3

Result files:

- `outputs/retrieval/results.csv`
- `outputs/retrieval/results_calm.csv`
- `outputs/retrieval/results_high_energy.csv`

Short discussion. The prototype supports interpretable filtering and quick experimentation. For broader narrative search, the next step is to add learned embeddings and a weighted ranking function.

7 Reproducibility and Artifacts

Key artifacts generated in this run:

- Classification metrics: `outputs/metrics/test_metrics.json`
- Classification report: `outputs/metrics/test_classification_report.json`
- ASR metrics: `outputs/metrics/asr_metrics.json`
- Figures: `outputs/metrics/figures/*.png`